

Natural Language Querying for Enterprise Data: Technology Overview

1. Executive Summary

Natural Language to SQL (NL-to-SQL) bridges business users and complex relational databases, empowering non-technical stakeholders to ask plain-English questions and receive immediate tabular and visual insights.

2. NL-to-SQL Architecture Components

Component	Function	Key Technology
Schema Metadata Ingestor	Extracts table schemas, column types, and foreign key relationships	SQLAlchemy Inspector
Contextual Vector Store	Indexes domain-specific business terms and KPI definitions	ChromaDB / FAISS
SQL Generation Engine	Translates NL queries into optimized, dialect-specific SQL queries	GPT-4 / LangChain SQL Agent
Query Validator & Sandbox	Strips destructive commands (DROP, DELETE, UPDATE) and checks syntax	SQLGlue + Read-Only DB Role
Insights & Chart Generator	Computes aggregations and generates executive narrative summaries	Pandas + Matplotlib / Seaborn

3. Security & Safety Principles

- Database connections must be strictly read-only (**SELECT** only).
- Query timeouts and row limit caps (**LIMIT 1000**) enforced on all generated queries.